

Kyle Tran

Mechanical Engineering, Robotics minor · Northeastern '29 · Seeking a Summer 2027 hardware or robotics co-op
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EDUCATION

Northeastern University — *Boston, MA*

Expected May 2029

B.S. Mechanical Engineering, Minor in Robotics · GPA 3.87 / 4.00

Coursework: Cornerstone of Engineering I, Statics, Thermodynamics, Intro to Materials Science, Probability and Statistics

PROJECTS

D.R.E.A.M.S — Autonomous Environmental Survey Boat — *personal project*

Jan 2026 – Present

- Designed a twin-hull catamaran in SolidWorks, concept to build: 3D-printed ASA hull, fiberglass-and-epoxy skin
- v1 completed autonomous waypoint surveys on local lakes: three RP2040 boards on a 433 MHz HC-12 radio link, and a winch that lowers a sensor pod to sample the water
- Designing the v2 KiCad carrier board (second revision, in bring-up) that replaces v1's hand-wired harness with three isolated battery domains, two VESC motor controllers and every sensor header
- Specified the v2 architecture (Raspberry Pi 5 plus two RP2040s, stereo vision, 4G telemetry) and lead its integration and testing; the software is written with AI coding agents

Combat Robotics — Team Dihydrogen Monoxide

Mar 2023 – Jun 2025

- Designed and built about ten robots across five weight classes in Onshape: spinner and drum weapons, brushless drive, aluminum frames sent out for CNC machining, 3D-printed nylon and polycarbonate armor
- 4th nationally in Plastic Antweight (2024 RCL National Championship); U.S. rankings of top-15 (1 lb), top-30 (1 lb plastic) and top-35 (3 lb); 4 first- and 8 second-place regional finishes
- Designed a single-cam shuffler drive so the newest version of Achilles (1 lb) walks instead of rolling

Autonomous Maze-Navigating Robot — *Cornerstone of Engineering I, team of two*

Oct 2025 – Dec 2025

- Designed a 3D-printed tank-drive chassis in Onshape to replace the class kit, which tipped in fast turns: 1.6 in tall, about half the kit's height, and every printed part fit the first time
- Wrote the MicroPython firmware: proportional wall-centering off two angled ultrasonic sensors, an LCD program menu, and asyncio so the LED and LCD animations never stall the motors
- Added crowned wheels to keep the rubber-band treads on in hard turns; completed all three challenge courses

Adaptive Compliant Robotic Arm — *personal project, in design*

2026 – Present

- Five-axis arm; four joints run 17:1 cycloidal gearboxes with output-side encoders, designed to hold $\pm 0.25^\circ$

LEADERSHIP & ACTIVITIES

President — *Combat Robots Club, Burlingame, CA*

Oct 2023 – Jun 2025

- Taught CAD, soldering, electronics and 3D printing; secured sponsorships and ran club finances and event travel
- Led the club to 16 podium finishes (4 first, 7 second, 5 third) across 5 events

Subsystem Lead — *Iron Panthers, FRC Team 5026, Burlingame, CA*

Sep 2021 – May 2024

- Designed and prototyped the spindexter mechanism in CAD; led 10+ students, advised by 5 professional mentors

Ceramic Tools R&D — *Burlingame High School*

Oct 2023 – May 2025

- Designed, 3D-printed and published 10+ tools for the ceramics classes, free for anyone to print on MakerWorld

SKILLS

CAD and design: SolidWorks, Onshape, Fusion 360, KiCad; mechanism design, design for manufacturing

Fabrication: 2.5D CNC machining, FDM 3D printing, composite layup (fiberglass, carbon fiber), laser cutting, soldering

Electronics: PCB design, Raspberry Pi, RP2040, MicroPython with asyncio, motor and sensor integration

Programming: Python, MATLAB, Java; AI coding agents (Claude Code)

AWARDS

4th, Robot Combat League National Championship, Plastic Antweight (2024) · 1st, CROS March Madness (2024) ·
1st, BCRC Bell Brawls IV (2024) · 1st, Spartan Bell Slugfest (2024) · 1st, NorCal Robotics Expo (2023)